

Bangladesh University of Engineering and Technology

Department of Computer Science & Engineering

MATH 141 Question Bank

Differential Calculus · Integral Calculus · ODE

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Table of Contents

Part I — Differential Calculus

- ▷ **S1.** Differential Calculus
 - Continuity and Differentiability
 - Successive Differentiation & Leibnitz’s Theorem
 - Maxima and Minima: Theoretical & Graphing Tasks
 - Applications of Maxima and Minima
 - Rolle’s Theorem & Mean Value Theorem
 - L’Hôpital’s Rule
 - Taylor’s & Maclaurin’s Theorems
 - Partial Differentiation & Euler’s Theorem
 - Tangent and Normal
 - Asymptotes
 - Rate of Change

Part II — Integral Calculus

- ▷ **S2.** Integral Calculus
 - Indefinite Integrals
 - Definite Integrals: Riemann Sums & Limits
 - Definite Integrals: Application of Properties
 - Definite Integrals: Substitution & Standard Methods
 - Reduction Formula & Wallis’ Formula
 - Improper Integrals
 - Beta & Gamma Functions
 - Area Under Curves
 - Arc Lengths
 - Volume & Surface Area of Revolution
 - Multiple Integrals

Part III — Ordinary Differential Equations

- ▷ **S3.** Ordinary Differential Equations
 - Formation of Differential Equations
 - First Order ODEs
 - Equations of First Order but Higher Degree
 - Second & Higher Order Linear ODEs
 - Euler’s Homogeneous Linear Equations
 - Applications of ODEs

BUET · MATH 141 Question Bank

Differential Calculus

Continuity, Differentiation, Maxima/Minima, Series Expansion & more

S1. Differential Calculus

Continuity and Differentiability

Q1. Define the limit of a function. Given

$$f(x) = \begin{cases} a + bx, & x < 1 \\ 4, & x = 1 \\ b - ax, & x > 1 \end{cases}$$

and $\lim_{x \rightarrow 1} f(x) = 1$, find all possible values of a and b . Also sketch the function $f(x)$ and discuss the continuity and differentiability of $f(x)$ at $x = 1$. **(13 marks)** [MATH 141 | L-1/T-1 | 2024–25]

Q2. Find the values of the constants a and b , if possible, that will make the function

$$f(x) = \begin{cases} \frac{x^4 + 1}{x^2}, & x < -2 \\ \frac{a(x+1)^3}{2}, & -2 \leq x \leq -\frac{1}{2} \\ x^2 + b, & x > -\frac{1}{2} \end{cases}$$

continuous everywhere. Also sketch and discuss differentiability at $x = -2$ and $x = -\frac{1}{2}$. **(18 marks)** [MATH 141 | L-1/T-1 | 2023–24]

Q3. Find the values of the constants k and m , if possible, that will make the function

$$f(x) = \begin{cases} 2x^3 + x + 7, & x \leq -1 \\ m(x+1) + k, & -1 < x \leq 2 \\ x^2 + 5, & x > 2 \end{cases}$$

continuous everywhere. Also discuss differentiability of $f(x)$ at $x = 2$. **(15 marks)** [MATH 141 | L-1/T-1 | 2022–23]

Q4. A function $f(x)$ is defined as:

$$f(x) = \begin{cases} 5x - 4, & x \leq 1 \\ cx^2 - dx, & 1 < x < 2 \\ 3x + 4, & x \geq 2 \end{cases}$$

Find the values of c and d for which $f(x)$ is continuous everywhere. Also discuss differentiability at $x = 1$ and $x = 2$. **(15 marks)** [MATH 141 | L-1/T-1 | 2021–22]

Q5. Sketch the graph of $f(x) = \begin{cases} 1 - x^2, & x < 0 \\ 1, & 0 \leq x < 1 \\ \frac{1}{x}, & x > 1 \end{cases}$ and discuss continuity and differentiability at $x = 0$ and $x = 1$. **(25 marks)** [MATH 145 | L-1/T-1 | 2020–21]

Q6. Suppose $f(x) = \begin{cases} s(x), & -\infty < x \leq \pi \\ \frac{\pi^2}{4} + 1, & \pi < x \leq 2\pi \\ t(x), & 2\pi < x < \infty \end{cases}$ where $y = s(x)$ is found by translating $y = x^2$ by $\frac{\pi}{2}$ units right and 1 unit upward; and

$y = t(x)$ is generated by (I) translating $y = |x|$ by 2π units right, (II) reflecting about the x -axis, (III) moving $\frac{\pi^2}{4}$ units upward. Sketch the graph of $f(x)$ and check differentiability at $x = \pi$. **(15 marks)** [MATH 145 | L-1/T-2 | 2019–20]

Q7. A function is defined as:

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 \leq x \leq 2 \\ \frac{1}{2}x^2, & x > 2 \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x = 1$ and $x = 2$. Also sketch the graph. **(18 marks)** [MATH 145 | L-1/T-1 | 2018–19]

Q8. A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} x^2 + 1, & 0 \leq x < 1/2 \\ 0, & x = 1/2 \\ x + 3, & 1/2 < x \leq 1 \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x = 0$ and $x = 1/2$. Also sketch the curve. **(12 marks)** [MATH 145 | L-1/T-1 | 2017–18]

- Q9.** Let $f(x) = \begin{cases} 4 + x^2, & 0 < x \leq 4 \\ 4, & -1 \leq x \leq 0 \\ 1 + x, & -4 \leq x < -1 \end{cases}$ Discuss the continuity and differentiability of $f(x)$ at $x = 0$ and $x = -1$. Also sketch the curve. **(12 marks)** [MATH 145 | L-1/T-1 | 2016–17]

- Q10.** Let $f(x) = \begin{cases} x^2 + x + a, & x \leq 0 \\ bx + 2, & x > 0 \end{cases}$ where a and b are constants. Find all values of a and b for which $f(x)$ is differentiable. **(12 marks)** [MATH 145 | L-1/T-1 | 2015–16]

- Q11.** A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} 1, & -\infty < x < 0 \\ 1 + \sin x, & 0 \leq x < \pi/2 \\ 2 + (x - \pi/2)^2, & \pi/2 \leq x < \infty \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x = 0$ and $x = \pi/2$. **(12 marks)** [MATH 141 | L-1/T-1 | 2014–15, 2006–07]

- Q12.** A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} 5x - 4, & 0 < x \leq 1 \\ 4x^2 - 3x, & 1 < x < 2 \\ 3x + 4, & x \geq 2 \end{cases}$$

Discuss the continuity of $f(x)$ at $x = 1$ and the existence of $f'(x)$ at $x = 2$. Also sketch the graph of $f(x)$. **(15 marks)** [MATH 141 | L-1/T-1 | 2013–14, 2005–06]

- Q13.** Is $f(x)$ defined below continuous at $x = 1$ and $x = 2$? Does $f'(x)$ exist for these values?

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 \leq x \leq 2 \\ x - \frac{1}{2}x^2, & x > 2 \end{cases}$$

Also sketch $f(x)$. **(20 marks)**

[MATH 141 | L-1/T-1 | 2012–13]

- Q14.** If $f(x) = \begin{cases} \frac{|x-2|}{x-2}, & x \neq 2 \\ 0, & x = 2 \end{cases}$, sketch the function $f(x)$ and find $\lim_{x \rightarrow 2} f(x)$. **(15 marks)** [MATH 141 | L-1/T-1 | 2012–13]

- Q15.** A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} 1 + x, & x \leq 0 \\ x, & 0 < x < 1 \\ 2 - x, & 1 \leq x \leq 2 \\ 2x - x^2, & x > 2 \end{cases}$$

Sketch the graph of $f(x)$ and discuss the continuity and differentiability of $f(x)$ at $x = 1$ and $x = 2$. **(25 marks)** [MATH 141 | L-1/T-1 | 2011–12]

- Q16.** Sketch the graph of the function

$$f(x) = \begin{cases} x^2 + 1, & x < 0 \\ 1, & 0 \leq x < 1 \\ \frac{1}{x}, & x \geq 1 \end{cases}$$

and discuss continuity and differentiability of $f(x)$ at $x = 0$ and at $x = 1$. **(25 marks)**

[MATH 141 | L-1/T-1 | 2010–11]

- Q17.** A function $f(x)$ is defined as:

$$f(x) = \begin{cases} -x, & x \leq 0 \\ x, & 0 < x < 1 \\ 2 - x, & x \geq 1 \end{cases}$$

Discuss the continuity of $f(x)$ at $x = 1$ and differentiability at $x = 0$. Also sketch the graph of $f(x)$. **(17 marks)** [MATH 141 | L-1/T-1 | 2008–09]

- Q18.** Show that every differentiable function must be continuous. Discuss the continuity of the following function:

$$f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

at $x = 0$. **(12 marks)**

[MATH 141 | L-1/T-1 | 2007–08]

Q19. A function $f(x)$ is defined as:

$$f(x) = \begin{cases} x^2 + 1 & x < 0 \\ 1 & 0 \leq x < 1 \\ x + 1 & x \geq 1 \end{cases}$$

Sketch the graph of $f(x)$. Find its domain and range. Test for continuity and differentiability at $x = 0$ and $x = 1$. **(18 marks)**
[MATH 141 | L-1/T-1 | 2004–05]

Q20. Prove that if a function $f(x)$ is differentiable at $x = x_0$, then it is also continuous at x_0 . Explain with an example. **(15 marks)**
[MATH 141 | L-1/T-1 | 2003–04]

Successive Differentiation & Leibnitz's Theorem

Q1. If $\cos^{-1}\left(\frac{y}{a}\right) = \ln\left(\frac{x}{n}\right)^n$, use Leibnitz's theorem to prove that

$$x^2 y_{n+2} + (2n+1)x y_{n+1} + 2n^2 y_n = 0.$$

(12 marks)

[MATH 141 | L-1/T-1 | 2024–25]

Q2. State Leibnitz's theorem. If $y = a \cos(\ln x) + b \sin(\ln x)$, show that $x^2 y_{n+2} + (2n+1)x y_{n+1} + (n^2+1)y_n = 0$. **(11 marks)** [MATH 141 | L-1/T-1 | 2023–24, 2019–20]

Q3. For $y = \frac{1}{2}(\tan^{-1} x)^2$, prove that $y_{n+2}(0) + 2n^2 y_n(0) + n(n-1)^2(n-2)y_{n-2}(0) = 0$. **(12 marks)** [MATH 141 | L-1/T-1 | 2022–23, 2012–13]

Q4. State Leibnitz's theorem. If $y = \sin(a \ln(x+b))$, prove that $(b+x)^2 y_{n+2} + (2n+1)(x+b)y_{n+1} + (n^2+a^2)y_n = 0$. **(15 marks)** [MATH 141 | L-1/T-1 | 2021–22]

Q5. If $y = e^{a \sin^{-1} x}$, show that $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2+a^2)y_n = 0$ and find y_n at $x = 0$. **(20 marks)** [MATH 145 | L-1/T-1 | 2020–21, 2016–17]
[MATH 141 | L-1/T-1 | 2007–08, 2004–05]

Q6. State and prove Leibnitz's theorem. If $y = \sin^{-1} x$, find y_n for $x = 0$. **(17 marks)** [MATH 145 | L-1/T-1 | 2018–19]

Q7. If $x = \tan u$, find $\{(1+x^2)u_{n+2} + 2(n+1)x u_{n+1} + n(n+1)u_n\}$ and hence find $(u_n)_0$. **(13 marks)** [MATH 145 | L-1/T-1 | 2017–18]

Q8. State Leibnitz's theorem. If $y^{1/m} + y^{-1/m} = 2x$, show that $(x^2-1)y_{n+2} + (2n+1)x y_{n+1} + (n^2-m^2)y_n = 0$. **(12 marks)** [MATH 145 | L-1/T-1 | 2015–16, 2006–07]

Q9. Use Leibnitz's theorem to find $y_n(x)$ at $x = 0$ both for even and odd n , where $y = (x + \sqrt{1+x^2})^m$. **(11 marks)** [MATH 141 | L-1/T-1 | 2014–15]

Q10. Find the n th derivative of $y = \tan^{-1} \frac{1+x}{1-x}$. **(10 marks)** [MATH 141 | L-1/T-1 | 2013–14]

Q11. Use Leibnitz's theorem to find the value of $\{1 - (ax+b)^2\}y_{n+2} - (2n+1)a(ax+b)y_{n+1} + (m^2-n^2)a^2 y_n$, if $y = A \sin\{m \sin^{-1}(ax+b)\}$. **(12 marks)** [MATH 141 | L-1/T-1 | 2013–14]

Q12. If $y = \sin(a \sin^{-1} x)$, show that $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2-a^2)y_n = 0$ and find the value of $(y_n)_0$. **(13 marks)** [MATH 141 | L-1/T-1 | 2011–12]

Q13. If $y = e^{m \cos^{-1} x}$, show that $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2+m^2)y_n = 0$ and find the value of y_n at $x = 0$. **(20 marks)** [MATH 141 | L-1/T-1 | 2010–11, 2004–05]

Q14. If $y = \frac{1}{x^2+a^2}$, show that $y_n = \frac{(-1)^n n!}{a^{n+2}} (\sin \theta)^{n+1} \sin(n+1)\theta$ where $\theta = \tan^{-1} \frac{a}{x}$. Also find the n th derivative of $y = \tan^{-1} x$ using this formula. **(18 marks)** [MATH 141 | L-1/T-1 | 2008–09]

Q15. If $y = (\sin^{-1} x)^2$, prove that $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - n^2 y_n = 0$ and hence find $(y_n)_0$. **(17 marks)** [MATH 141 | L-1/T-1 | 2008–09]

Q16. Find the n th derivative of $y = x^3 e^x \cos^2 x$. **(11 marks)** [MATH 141 | L-1/T-1 | 2006–07]

Q17. If $y = [x - \sqrt{1+x^2}]^m$, prove that $(1+x^2)y_{n+2} + (2n+1)x y_{n+1} + (n^2-m^2)y_n = 0$ and find y_n at $x = 0$. **(17 marks)** [MATH 141 | L-1/T-1 | 2005–06]

Q18. If $y = e^{a \sin^{-1} x} = a_0 + a_1 x + a_2 x^2 + \dots$, show that:

(a) $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2+a^2)y_n = 0$

(b) $(n+1)(n+2)a_{n+2} = (n^2 + a^2)a_n$

(16 marks)

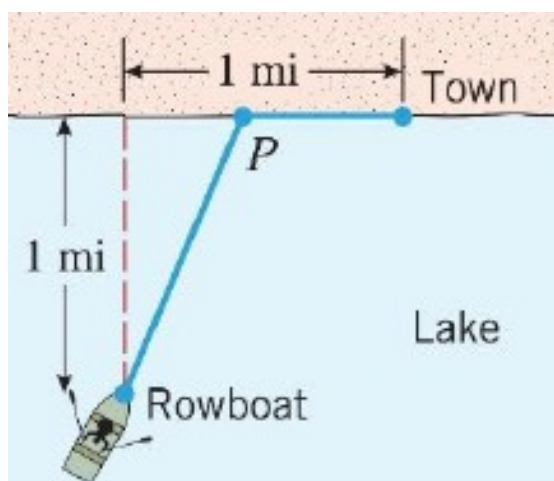
[MATH 141 | L-1/T-1 | 2003–04]

Maxima and Minima: Theoretical & Graphing Tasks

- Q1.** “The point of inflection is independent of the first derivative.” Justify your answer with an example. (10 marks) [MATH 141 | L-1/T-1 | 2024–25]
- Q2.** Locate the relative maximum and minimum of $f(x) = \frac{4}{x} + \frac{36}{2-x}$. Also find the extremum value. (12 marks) [MATH 141 | L-1/T-1 | 2024–25]
- Q3.** State and prove the necessary conditions for the existence of extremum values. (8 marks) [MATH 141 | L-1/T-1 | 2023–24]
- Q4.** If $f(x) = ax^3 + bx^2 + cx + d$ ($a \neq 0$): (i) Prove that the cubic polynomial $f(x)$ has exactly one inflection point. (ii) Prove that if a cubic polynomial has three x -intercepts, then the inflection point occurs at the average value of the intercepts. (iii) Use (ii) to find the inflection point of $f(x) = x^3 - 3x^2 + 2x$ and check using f'' . (15 marks) [MATH 141 | L-1/T-1 | 2021–22]
- Q5.** For the function $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 5$, discuss maximum and minimum and concavity, then sketch the graph. (15 marks) [MATH 145 | L-1/T-1 | 2020–21] [MATH 141 | L-1/T-1 | 2010–11]
- Q6.** Sketch the graph of $y = x^3 - 3x + 2$ and identify the locations of intercepts, relative extrema, and inflection points. Also find the intervals where $f(x)$ is increasing or decreasing. (12 marks) [MATH 145 | L-1/T-1 | 2016–17]
- Q7.** If $y = (2x+3)^2(3x-1)^2$, determine the nature of the stationary points. (12 marks) [MATH 141 | L-1/T-1 | 2011–12]
- Q8.** Find the intervals on which the function $f(x) = x^3 - 12x + 4$ is concave up and concave down. (8 marks) [MATH 141 | L-1/T-1 | 2011–12]
- Q9.** Find the maximum and minimum values of y if $\frac{dy}{dx} = (x-a)^4(x-b)^3$. (17 marks) [MATH 141 | L-1/T-1 | 2006–07]
- Q10.** Discuss the maxima and minima of $f(x) = x^5 - 5x^4 + 5x^3 - 1$. Sketch the graph indicating points of inflexion. (20 marks) [MATH 141 | L-1/T-1 | 2004–05]

Applications of Maxima and Minima

- Q1.** The population of bacteria is governed by $p(t) = \frac{P_0 k e^{kt}}{k - P_0 + P_0 e^{kt}}$, where P_0 is the initial bacteria. Find the maximum amounts of bacteria in the medium and the time for fastest growth. (17 marks) [MATH 141 | L-1/T-1 | 2023–24]
- Q2.** A metal can with a volume of 60 in^3 is to be constructed as a right circular cylinder. If the cost of the side material is $\$0.05/\text{in}^2$ and the cost for the top and bottom is $\$0.03/\text{in}^2$, find the dimensions that minimise the cost. (13 marks) [MATH 141 | L-1/T-1 | 2022–23]
- Q3.** A man is floating in a rowboat 1 mile from the straight shoreline of a large lake. A town is located on the shoreline 1 mile from the nearest point on the shore. He rows to a point P on the shoreline and then walks to the town. To what point should he row to reach the town in least time if he can walk at 4 mi/h and row at 2 mi/h?



(15 marks)

[MATH 145 | L-1/T-2 | 2019–20]

- Q4.** Show that of all rectangles of given area, the square has the smallest perimeter. (10 marks) [MATH 145 | L-1/T-1 | 2018–19]

- Q5. Find the largest rectangle that can be inscribed within the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. (17 marks) [MATH 145 | L-1/T-1 | 2017–18]
- Q6. An open box is to be made from a 16-inch by 30-inch piece of cardboard by cutting out squares of equal size from the four corners and bending up the sides. What size should the square be to obtain a box with the largest volume? (12 marks) [MATH 145 | L-1/T-1 | 2015–16]
- Q7. Find the volume of the greatest cylinder that can be inscribed in a right circular cone of height h and semi-vertical angle α . (11 marks) [MATH 141 | L-1/T-1 | 2014–15]
- Q8. Find the radius and height of the right circular cylinder of largest volume that can be inscribed in a sphere of radius 6 inches. (12 marks) [MATH 141 | L-1/T-1 | 2013–14]
- Q9. A rectangular warehouse with a flat roof is to have a floor area of 9600 square feet. The interior is to be divided into store room and office space by an interior wall parallel to one pair of the sides. The roof and the floor area are same. Find the dimensions that minimise the total length of wall. (20 marks) [MATH 141 | L-1/T-1 | 2012–13]
- Q10. Find the volume of the greatest right circular cone that can be inscribed in a given sphere of radius a . (10 marks) [MATH 141 | L-1/T-1 | 2008–09]
- Q11. If 40 square metres of sheet is to be used in the construction of an open tank with a square base, find the dimensions so that the capacity (volume) is the greatest. (13 marks) [MATH 141 | L-1/T-1 | 2007–08]
- Q12. Prove that the height of the cylinder of greatest volume inscribed in a sphere of radius a is $\frac{2\sqrt{3}}{3}a$. (18 marks) [MATH 141 | L-1/T-1 | 2005–06]
- Q13. Show that the radius of the right circular cylinder of greatest curved surface inscribed in a given cone is half that of the base of the cone. (15 marks) [MATH 141 | L-1/T-1 | 2004–05]
- Q14. Find the volume of the largest possible right circular cylinder that can be inscribed in a sphere of radius a . (17 marks) [MATH 141 | L-1/T-1 | 2003–04]

Rolle's Theorem & Mean Value Theorem

- Q1. Explain the velocity interpretation of the Mean Value Theorem. Verify the Mean Value Theorem for $f(x) = \tan x$ on the interval $[0, \pi]$. (10 marks) [MATH 141 | L-1/T-1 | 2024–25]
- Q2. State and prove Rolle's theorem. (10 marks) [MATH 141 | L-1/T-1 | 2023–24]
- Q3. State/Discuss/Verify Rolle's theorem for $f(x) = \ln \left[\frac{x^2 + ab}{(a+b)x} \right]$ on/in $[a, b]$ or (a, b) , where a and b are positive constants. (10 marks) [MATH 141 | L-1/T-1 | 2022–23, 2007–08] [MATH 145 | L-1/T-1 | 2016–17]
- Q4. In the Mean Value Theorem $f(a+h) - f(a) = hf'(a+\theta h)$, $0 < \theta < 1$, if $f(x) = \frac{1}{3}x^3 - \frac{3}{2}x^2 + 2x$ and $a = 0$, $h = 3$, show that θ has two values and find them. (10 marks) [MATH 141 | L-1/T-1 | 2022–23, 2013–14, 2005–06]
- Q5. State Mean Value Theorem. Verify the Mean Value Theorem for $f(x) = x^{2/3}$ on the interval $[-2, 3]$. (10 marks) [MATH 141 | L-1/T-1 | 2021–22]
- Q6. State and prove Rolle's theorem. Find the two x -intercepts of $f(x) = x^2 - 5x + 4$ and confirm that $f'(c) = 0$ at some point c between those intercepts. Sketch the graph. (17 marks) [MATH 145 | L-1/T-1 | 2020–21]
- Q7. State and prove Rolle's theorem. Suppose two runners in a 100 m dash finish in a tie. Show that they had the same velocity at least once during the race. (18 marks) [MATH 145 | L-1/T-1 | 2018–19]
- Q8. If the Mean Value Theorem is $f(b) - f(a) = (b-a)f'(x_1)$, find x_1 where $f(x) = x^2 - 3x - 1$, $a = 11/7$, and $b = 13/7$. (11 marks) [MATH 145 | L-1/T-1 | 2017–18]
- Q9. Verify Cauchy's mean value theorem for $f(x) = x^2 - 2x + 3$ and $g(x) = x^3 - 7x^2 + 26x - 5$ in the interval $[-1, 1]$. (11 marks) [MATH 145 | L-1/T-1 | 2015–16]
- Q10. State Mean Value Theorem. If $f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x+\theta h)$, $0 < \theta < 1$, find the value of θ for $x = a$ when $f(x) = (x-a)^{5/2}$. (12 marks) [MATH 141 | L-1/T-1 | 2014–15]
- Q11. State and prove the first mean value theorem. Find the value of c (critical value) using Cauchy's mean value theorem for $f(x) = e^x$ and $g(x) = e^{-x}$ in the interval $(3, 5)$. (15 marks) [MATH 141 | L-1/T-1 | 2012–13]

- Q12.** State and prove Rolle's Theorem. Is Rolle's theorem applicable to the function $f(x) = \begin{cases} x, & 0 \leq x < 1 \\ -x, & -1 \leq x \leq 0 \end{cases}$ in $[-1, 1]$? **(15 marks)**
[MATH 141 | L-1/T-1 | 2011–12]
- Q13.** Verify the Mean Value Theorem for the function $f(x) = Ax^2 + Bx + C$ in the interval (a, b) . **(10 marks)** [MATH 141 | L-1/T-1 | 2011–12]
- Q14.** State and prove Rolle's theorem and verify it for the function $f(x) = x^3 - x^2 - 4x + 4$ in the interval $(-2, 2)$. **(17 marks)** [MATH 141 | L-1/T-1 | 2010–11, 2006–07]
- Q15.** State and verify the mean value theorem for $f(x) = 2x^2 - 7x + 10$ in the interval $[2, 5]$. **(10 marks)** [MATH 141 | L-1/T-1 | 2008–09]
- Q16.** State and prove Rolle's theorem. Verify the mean value theorem for $f(x) = 3 + 2x - x^2$ in the interval $[0, 1]$. **(17 marks)** [MATH 141 | L-1/T-1 | 2004–05]
- Q17.** State and prove the first mean value theorem. If $f(h) = f(0) + hf'(0) + \frac{h^2}{2}f''(\theta h)$, $0 < \theta < 1$, find θ when $h = 7$ and $f(x) = \frac{1}{1+x}$. **(20 marks)**
[MATH 141 | L-1/T-1 | 2003–04]

L'Hôpital's Rule

- Q1.** Applying the hypothesis of L'Hôpital's theorem, evaluate the limit
$$\lim_{x \rightarrow +\infty} a^x \sin\left(\frac{b}{a^x}\right), \quad a > 1.$$
(10 marks) [MATH 141 | L-1/T-1 | 2024–25]
- Q2.** Evaluate $\lim_{x \rightarrow 2} \left(\frac{1}{x-2} - \frac{1}{\ln(x-1)} \right)$. **(6 marks)** [MATH 141 | L-1/T-1 | 2023–24]
- Q3.** Evaluate $\lim_{x \rightarrow 0} (1 - x^2)^{1/\log(1-x^2)}$. **(8 marks)** [MATH 141 | L-1/T-1 | 2022–23]
- Q4.** Find $\lim_{x \rightarrow +\infty} [\ln x - \ln(1+x)]$. **(5 marks)** [MATH 141 | L-1/T-1 | 2021–22]
- Q5.** Test whether $\lim_{x \rightarrow \pi/2} \frac{e^{\tan x} - 1}{e^{\tan x} + 1}$ exists. If so, find its value. **(10 marks)** [MATH 145 | L-1/T-1 | 2020–21]
- Q6.** Evaluate $\lim_{x \rightarrow \infty} \left(\frac{c_1^{1/x} + c_2^{1/x} + c_3^{1/x} + \dots + c_n^{1/x}}{n} \right)^{nx}$. **(15 marks)** [MATH 145 | L-1/T-2 | 2019–20]
- Q7.** Evaluate $\lim_{x \rightarrow 0} x^{2 \sin x}$. **(7 marks)** [MATH 145 | L-1/T-1 | 2018–19]
- Q8.** Evaluate $\lim_{x \rightarrow \infty} (x + e^x)^{1/x}$. **(10 marks)** [MATH 145 | L-1/T-1 | 2017–18]
- Q9.** Evaluate $\lim_{x \rightarrow 1} (1 - x^2)^{1/\ln(1-x)}$. **(11 marks)** [MATH 145 | L-1/T-1 | 2016–17]
- Q10.** Evaluate $\lim_{x \rightarrow 0} x \ln(\sin x)$. **(11 marks)** [MATH 145 | L-1/T-1 | 2015–16]
- Q11.** Evaluate: (i) $\lim_{x \rightarrow \pi} \left(\frac{2\pi}{x} - 1 \right)^{\tan(x/2)}$ (ii) $\lim_{x \rightarrow 0} \cot x \cdot \log \frac{1-x}{1+x}$. **(12 marks)** [MATH 141 | L-1/T-1 | 2014–15]
- Q12.** Evaluate $\lim_{x \rightarrow 1} (1 - x^2)^{1/\log_e(1-x)}$. **(10 marks)** [MATH 141 | L-1/T-1 | 2013–14]
- Q13.** Find $\lim_{x \rightarrow 0} (\cos x)^{1/x^2}$. **(10 marks)** [MATH 141 | L-1/T-1 | 2011–12]
- Q14.** Test whether the limit $\lim_{x \rightarrow \pi/2} \frac{e^{\tan x} - 1}{e^{\tan x} + 1}$ exists. If possible, find the value of the limit. **(10 marks)** [MATH 141 | L-1/T-1 | 2010–11]
- Q15.** Find the value of the series $\lim_{n \rightarrow \infty} \frac{1 + 2^{11} + 3^{11} + \dots + n^{11}}{n^{12}}$. **(10 marks)** [MATH 143 | L-1/T-2 | 2010–11]
- Q16.** Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$. **(10 marks)** [MATH 141 | L-1/T-1 | 2008–09]

- Q17. Evaluate: (i) $\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x}$ (ii) $\lim_{x \rightarrow 0} (a^x + x)^{1/x}$. (12 marks) [MATH 141 | L-1/T-1 | 2007–08]
- Q18. Evaluate $\lim_{x \rightarrow 0} \left[\frac{\log \log(1-x^2)}{\log \log \cos x} \right]$. (9 marks) [MATH 141 | L-1/T-1 | 2006–07]
- Q19. Evaluate $\lim_{x \rightarrow 0} \frac{x - \sin^{-1} x}{\sin^3 x}$. (8 marks) [MATH 141 | L-1/T-1 | 2005–06]
- Q20. Evaluate $\lim_{x \rightarrow 0} \left[\frac{1}{x} - \frac{1}{\sin x} \right]$. (7 marks) [MATH 141 | L-1/T-1 | 2003–04]

Taylor's & Maclaurin's Theorems

- Q1. Determine the Taylor series of $f(x) = \sin x$ around $x = \frac{\pi}{2}$ with Lagrange form of remainder after n terms, and calculate its radius of convergence. (12 marks) [MATH 141 | L-1/T-1 | 2024–25]
- Q2. Expand $\cos x$ in powers of $\left(x - \frac{\pi}{3}\right)$ with Lagrange's form of remainder, then approximate $\cos 61^\circ$ using the first seven terms and estimate the accuracy. (13 marks) [MATH 141 | L-1/T-1 | 2023–24]
- Q3. Find the Taylor polynomial $P_n(x)$ for $f(x) = e^x$ at $x = 0$. Hence compute $P_5(0.1)$ and compare with $f(0.1)$. Also compute the error bound for $P_5(0.1)$ using Lagrange's remainder. (10 marks) [MATH 141 | L-1/T-1 | 2021–22]
- Q4. Find the Lagrange's form of remainder after n terms in the expansion of $e^{ax} \cos bx$ in powers of x . (10 marks) [MATH 145 | L-1/T-1 | 2018–19]
- Q5. Expand $(\sin^{-1} x)^2$ in a series of ascending powers of x . (14 marks) [MATH 145 | L-1/T-1 | 2017–18]
- Q6. Expand $\ln x$ in powers of $(x - 2)$. Also find the Lagrange and Cauchy forms of remainder. (12 marks) [MATH 145 | L-1/T-1 | 2016–17]
- Q7. Expand the polynomial $2x^3 + 7x^2 + x - 1$ in powers of $(x - 2)$ and approximate it to five decimal-place accuracy. (12 marks) [MATH 145 | L-1/T-1 | 2015–16]
- Q8. Expand $y = \log \sec x$ in a series of ascending powers of x up to the sixth power of x . (12 marks) [MATH 141 | L-1/T-1 | 2014–15]
- Q9. Expand $y = e^x \log_e(1+x)$ in a series of ascending powers of x . (10 marks) [MATH 141 | L-1/T-1 | 2013–14]
- Q10. Find the infinite series of $y = \log(1 + \cos 3x)$ and state the condition under which the expansion is valid. (15 marks) [MATH 141 | L-1/T-1 | 2012–13]
- Q11. Find θ if $f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x+\theta h)$, ($0 < \theta < 1$), where $f(x) = ax^3 + bx^2 + cx + d$. (11 marks) [MATH 141 | L-1/T-1 | 2007–08]
- Q12. State and prove Taylor's theorem with remainder. (13 marks) [MATH 141 | L-1/T-1 | 2007–08]
- Q13. Apply Maclaurin's theorem to obtain terms up to x^4 in the expansion of $\log(1 + \sin^2 x)$. (10 marks) [MATH 141 | L-1/T-1 | 2006–07]

Partial Differentiation & Euler's Theorem

- Q1. If $u = f(x, y)$, $x = r \cosh \theta$, $y = r \sinh \theta$, show that

$$\left(\frac{\partial u}{\partial r}\right)^2 - \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta}\right)^2 = \left(\frac{\partial u}{\partial x}\right)^2 - \left(\frac{\partial u}{\partial y}\right)^2.$$

(13 marks)

[MATH 141 | L-1/T-1 | 2024–25]

- Q2. If $z = x \log(x+r) - r$ where $r^2 = x^2 + y^2$, show that (i) $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{1}{x+r}$ (ii) $\frac{\partial^3 z}{\partial x^3} = -\frac{x}{r^3}$. (12 marks) [MATH 141 | L-1/T-1 | 2023–24]
- Q3. If $V = V(r)$ and $r^2 = x^2 + y^2 + z^2$, show that $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = \frac{d^2 V}{dr^2} + \frac{2}{r} \frac{dV}{dr}$. (15 marks) [MATH 141 | L-1/T-1 | 2022–23, 2005–06]

- Q4.** State and prove Euler's theorem for homogeneous functions. Verify Euler's theorem for $f(x, y) = \frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}$. **(15 marks)** [MATH 141 | L-1/T-1 | 2021–22]
- Q5.** If $u = \tan^{-1} \frac{x^3 - y^3}{x + y}$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$ and evaluate $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy}$. **(18 marks)** [MATH 145 | L-1/T-1 | 2020–21]
- Q6.** Show that $x^2 \frac{\partial^2 u}{\partial x^2} + y^2 \frac{\partial^2 u}{\partial y^2} + z^2 \frac{\partial^2 u}{\partial z^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + 2yz \frac{\partial^2 u}{\partial y \partial z} + 2zx \frac{\partial^2 u}{\partial z \partial x} = n(n-1)u$, where $u = x^n f\left(\frac{y}{x}, \frac{z}{x}\right)$. **(15 marks)** [MATH 145 | L-1/T-2 | 2019–20]
- Q7.** If $u = x^2 \tan^{-1} \frac{x}{y} - y^2 \tan^{-1} \frac{y}{x}$, find the value of $\frac{\partial^2 u}{\partial x \partial y}$. **(10 marks)** [MATH 145 | L-1/T-1 | 2018–19]
- Q8.** Verify Euler's theorem for $u = \frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}}$. **(10 marks)** [MATH 145 | L-1/T-1 | 2017–18]
- Q9.** If $u = \ln(x^2 + y^2 + z^2)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 2$. **(11 marks)** [MATH 145 | L-1/T-1 | 2016–17]
- Q10.** If $u = f(y - z, z - x, x - y)$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. **(12 marks)** [MATH 145 | L-1/T-1 | 2015–16]
- Q11.** If $u = \cos^{-1} \frac{x + y}{\sqrt{x} + \sqrt{y}}$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0$. **(11 marks)** [MATH 145 | L-1/T-1 | 2015–16]
- Q12.** State Euler's theorem for homogeneous functions of two variables of degree n . If $u = \sin^{-1} \frac{x + y}{\sqrt{x} + \sqrt{y}}$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$. **(11 marks)** [MATH 141 | L-1/T-1 | 2014–15]
- Q13.** If $u = f(x^2 + 2yz, y^2 + 2zx, z^2 + 2xy)$, find the value of $(y^2 - zx) \frac{\partial u}{\partial x} + (x^2 - yz) \frac{\partial u}{\partial y} + (z^2 - xy) \frac{\partial u}{\partial z}$. **(12 marks)** [MATH 141 | L-1/T-1 | 2014–15]
- Q14.** Define homogeneous function. If $u = x/r^3$ and $r^2 = x^2 + y^2 + z^2$, find the value of $u_{xx} + u_{yy} + u_{zz}$. **(13 marks)** [MATH 141 | L-1/T-1 | 2013–14]
- Q15.** If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$ and evaluate $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy}$. **(18 marks)** [MATH 141 | L-1/T-1 | 2012–13, 2011–12, 2010–11]
- Q16.** If $u = \ln(x^3 + y^3 + z^3 - 3xyz)$, show that
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = -\frac{3}{(x + y + z)^2}.$$
 (15 marks) [MATH 141 | L-1/T-1 | 2008–09]
- Q17.** If $u = x^n f(y/x, z/x)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu$. **(12 marks)** [MATH 141 | L-1/T-1 | 2007–08]
- Q18.** If $Z = x^m f(y/x) + x^n g(y/x)$, show that:
$$x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} + mnz = (m + n - 1) \left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right).$$
 (18 marks) [MATH 141 | L-1/T-1 | 2006–07]
- Q19.** If $u = u(x, y)$, $x = r \cos \theta$, $y = r \sin \theta$, prove that:
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}.$$
 (18 marks) [MATH 141 | L-1/T-1 | 2004–05]
- Q20.** Transform the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ into polar form. **(12 marks)** [MATH 141 | L-1/T-1 | 2003–04]

- Q1.** Find the equation of the tangent and normal to the curve $y(x-2)(x-3) - x + 7 = 0$ at the point of x -interception. Also evaluate the subtangent, subnormal, length of tangent, and length of normal of the curve at the point. **(13 marks)** [MATH 141 | L-1/T-1 | 2024–25]
- Q2.** Find the equation of normal lines at the inflection points of the function $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 5$. **(10 marks)** [MATH 141 | L-1/T-1 | 2023–24]
- Q3.** Show that in the curve $a^2y^5 = k(bx+c)^4$, the cube of the subtangent varies as the fifth power of the subnormal. **(11 marks)** [MATH 141 | L-1/T-1 | 2022–23]
- Q4.** Find the equations of the tangent lines at the inflection points of $y = x^4 - 6x^3 + 12x^2 - 8x$. **(11 marks)** [MATH 141 | L-1/T-1 | 2022–23]
- Q5.** If the normal to the curve $x^{2/3} + y^{2/3} = a^{2/3}$ makes an angle ϕ with the axis of x , show that its equation is $y \cos \phi - x \sin \phi = a \cos 2\phi$. **(15 marks)** [MATH 145 | L-1/T-1 | 2020–21] [MATH 141 | L-1/T-1 | 2010–11]
- Q6.** If the normal to the curve $x^{2/3} + y^{2/3} = a^{2/3}$ makes an angle θ with the axis of x , find the value of $(y \cos \theta - x \sin \theta)$. **(18 marks)** [MATH 145 | L-1/T-1 | 2017–18]
- Q7.** Show that the condition that the curves $x^{2/3} + y^{2/3} = c^{2/3}$ may touch the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $a + b = c$. **(12 marks)** [MATH 145 | L-1/T-1 | 2016–17] [MATH 141 | L-1/T-1 | 2011–12]
- Q8.** In the curve $x = \frac{y^2 - a^2}{4a} - \frac{a}{2} \log \frac{y}{a}$, show that the difference between the lengths of the tangent and the sub-tangent is constant. **(12 marks)** [MATH 141 | L-1/T-1 | 2014–15]
- Q9.** If p_1 and p_2 be the perpendiculars from the origin to the tangent and normal respectively at any point (x, y) on the curve $x^{2/3} + y^{2/3} = a^{2/3}$, show that $4p_1^2 + p_2^2 = a^2$. **(12 marks)** [MATH 141 | L-1/T-1 | 2013–14]
- Q10.** Show that the curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ touches the straight line $\frac{x}{a} + \frac{y}{b} = 2$ at the point (a, b) whatever be the value of n . **(10 marks)** [MATH 141 | L-1/T-1 | 2012–13]
- Q11.** Find the condition that the two curves $\frac{x^2}{a} + \frac{y^2}{b} = 1$ and $\frac{x^2}{a'} + \frac{y^2}{b'} = 1$ will cut orthogonally. **(12 marks)** [MATH 141 | L-1/T-1 | 2008–09]
- Q12.** Find the angle of intersection between the curves $x^2 - y^2 = 2a^2$ and $x^2 + y^2 = 4a^2$. **(10 marks)** [MATH 141 | L-1/T-1 | 2007–08]
- Q13.** If $x \cos \alpha + y \sin \alpha = p$ touches the curve $\frac{x^m}{a^m} + \frac{y^m}{b^m} = 1$, show that $(a \cos \alpha)^{m/(m-1)} + (b \sin \alpha)^{m/(m-1)} = p^{m/(m-1)}$. **(11 marks)** [MATH 141 | L-1/T-1 | 2006–07]
- Q14.** If x_1, y_1 are the intercepts on the axes of x and y by the tangent at any point (x, y) to the curve $(x/a)^{2/3} + (y/b)^{2/3} = 1$, show that $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$. **(17 marks)** [MATH 141 | L-1/T-1 | 2005–06]
- Q15.** Show that for the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the radius of curvature at the end of the major axis equals half of the latus rectum. **(17 marks)** [MATH 141 | L-1/T-1 | 2005–06]
- Q16.** Prove that the segment between the coordinate axes of any tangent to the curve $x^{2/3} + y^{2/3} = a^{2/3}$ is of constant length. **(17 marks)** [MATH 141 | L-1/T-1 | 2004–05]
- Q17.** Show that the radii of curvature at the origin for the curve $x^3 + y^3 = 3axy$ are each equal to $\frac{3a}{2}$. **(18 marks)** [MATH 141 | L-1/T-1 | 2004–05]
- Q18.** Find the coordinates of the centre of curvature at any point of the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$. Hence show that the evolute of the cycloid is another cycloid. **(17 marks)** [MATH 141 | L-1/T-1 | 2003–04]

Asymptotes

- Q1.** Find the asymptotes of the hyperbola $x^2 - y^2 + 3x - 7y = 3$. **(12 marks)** [MATH 145 | L-1/T-1 | 2020–21]
- Q2.** Obtain the asymptotes of the hyperbola $8xy + 3x^2 - 3y^2 + 6x + 8y + 4 = 0$. Also find the equation of the conjugate hyperbola. **(17 marks)** [MATH 145 | L-1/T-1 | 2015–16]
- Q3.** Find all the asymptotes of the curve $x^3 - x^2y - xy^2 + y^3 + 2x^2 - 4y^2 + 2xy + x + y + 1 = 0$. **(12 marks)** [MATH 141 | L-1/T-1 | 2014–15, 2003–04]

- Q4.** Find all the asymptotes of the curve $x^3 + 3x^2y - xy^2 - 3y^3 + x^2 - 2xy + 3y^2 + 4x + 5 = 0$. **(13 marks)** [MATH 141 | L-1/T-1 | 2013–14, 2005–06]
- Q5.** Find the area of the region formed by the asymptotes of the curve $(x^2 - y^2)^2 = 2(x^2 + y^2)$. **(15 marks)** [MATH 141 | L-1/T-1 | 2012–13]
- Q6.** Find the area of the triangle formed by the asymptotes of the curve $(x^2 - y^2)y - 2ay^2 + 5x - 7 = 0$. **(15 marks)** [MATH 141 | L-1/T-1 | 2011–12]
- Q7.** Find the asymptotes of the curve $x^3 - 2y^3 + xy(2x - y) + y(x - y) + 1 = 0$. **(10 marks)** [MATH 141 | L-1/T-1 | 2010–11]
- Q8.** Find the asymptotes of the curve $x^2 - y^2 - 6x + 8y - 8 = 0$. **(11 marks)** [MATH 141 | L-1/T-1 | 2008–09]
- Q9.** Obtain all asymptotes of the curve $x^3 + 3x^2y - 4y^3 - x + y + 3 = 0$. **(15 marks)** [MATH 141 | L-1/T-1 | 2007–08]
- Q10.** Find all asymptotes of $y^3 - 5xy^2 + 8x^2y - 4x^3 - 4y^2 + 12xy - 8x^2 + 3y - 3x + 2 = 0$. **(13 marks)** [MATH 141 | L-1/T-1 | 2006–07]

Rate of Change

- Q1.** The volume of a spherical balloon is increasing at the rate of $10 \text{ cm}^3/\text{sec}$. Find the rate of change of its surface at the instant when its radius is 16 cm. **(8 marks)** [MATH 141 | L-1/T-1 | 2008–09]

BUET · MATH 141 Question Bank

Integral Calculus

Definite Integrals, Beta/Gamma, Multiple Integrals, Applications & more

S2. Integral Calculus

Indefinite Integrals

Q1. Evaluate $\int \frac{x^3}{(x-1)\sqrt{x-2}} dx$. **(15 marks)** [MATH 145 | L-1/T-2 | 2019–20]

Q2. Evaluate $\int \frac{2 \cos x + 3 \sin x}{4 \cos x - 5 \sin x} dx$. **(15 marks)** [MATH 145 | L-1/T-2 | 2019–20]

Q3. Compute $\int \frac{3x+2}{5x^2+2x+3} dx$. **(10 marks)** [MATH 143 | L-1/T-2 | 2014–15]

Q4. Work out the following:

(a) $\int \frac{(x+3) dx}{\sqrt{4x^2-4x+3}}$

(b) $\int \frac{dx}{\sin 2x + \sin x}$

(10+12 marks)

[MATH 143 | L-1/T-2 | 2013–14]

Q5. Work out the following:

(a) $\int \frac{dx}{\sqrt{x^2+4}(x^2+1)}$

(b) $\int \frac{\cos x dx}{5-3 \cos x}$

(12+11 marks)

[MATH 143 | L-1/T-2 | 2012–13]

Q6. Work out the following integrals:

(a) $\int \frac{x^3}{(2+3x^2)^{5/2}} dx$

(b) $\int \sin^2 x \cos^6 x dx$

[MATH 143 | L-1/T-2 | 2011–12]

Q7. Work out the following integrals:

(a) $\int \frac{dx}{x^{5/3}(2-x)^{1/3}}$

(b) $\int \frac{x dx}{(7x-10-x^2)^{3/2}}$

(12+12 marks)

[MATH 143 | L-1/T-2 | 2010–11]

Definite Integrals: Riemann Sums & Limits

Q1. Define the definite integral of a function and hence evaluate

$$\lim_{n \rightarrow \infty} \left[\frac{\sqrt{n}}{\sqrt{n^3}} + \frac{\sqrt{n}}{\sqrt{(n+4)^3}} + \frac{\sqrt{n}}{\sqrt{(n+8)^3}} + \cdots + \frac{\sqrt{n}}{\sqrt{(n+4(n-1))^3}} \right]$$

using the properties of definite integrals. **(13 marks)**

[MATH 141 | L-1/T-1 | 2024–25]

Q2. Evaluate $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1^2}{n^2}\right)^{1/n^2} \left(1 + \frac{4}{n^2}\right)^{2/n^2} \cdots \left(1 + \frac{n^2}{n^2}\right)^{2n/n^2} \right]$ using properties of definite integrals. **(10 marks)** [MATH 141 | L-1/T-1 | 2023–24]

Q3. Find $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1^2}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \left(1 + \frac{3^2}{n^2}\right) \cdots \left(1 + \frac{n^2}{n^2}\right) \right]^{1/n}$ using an integral formula. **(11 marks)** [MATH 141 | L-1/T-1 | 2022–23] [MATH 143 | L-1/T-2 | 2014–15]

Q4. Find the area under the parabola $f(x) = 9 - x^2$ over $[0, 3]$ using summation of series with midpoint values. **(10 marks)** [MATH 145 | L-1/T-1 | 2020–21]

Q5. Evaluate $\lim_{n \rightarrow \infty} \left\{ \left(1 + \frac{1^2}{n^2}\right)^{2/n} \left(1 + \frac{2^2}{n^2}\right)^{4/n} \cdots \left(1 + \frac{n^2}{n^2}\right)^{2n/n} \right\}$. **(12 marks)** [MATH 145 | L-1/T-1 | 2018–19]

Q6. Find the value of the following series: $\lim_{n \rightarrow \infty} \left[\frac{n}{n^2 + 1^2} + \frac{n}{n^2 + 2^2} + \frac{n}{n^2 + 3^2} + \cdots + \frac{n}{n^2 + n^2} \right]$. **(11 marks)** [MATH 145 | L-1/T-1 | 2017–18]

Q7. Evaluate $\lim_{n \rightarrow \infty} \frac{1}{n} \left[\sin^{2k} \frac{\pi}{2n} + \sin^{2k} \frac{2\pi}{2n} + \cdots + \sin^{2k} \frac{\pi}{2} \right]$. **(12 marks)** [MATH 145 | L-1/T-1 | 2015–16]

Definite Integrals: Application of Properties

Q1. Evaluate $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$ using suitable properties of definite integrals. **(10 marks)** [MATH 141 | L-1/T-1 | 2021–22] [MATH 143 | L-1/T-2 | 2014–15]

Q2. Evaluate $\int_0^{\pi/2} \frac{x dx}{\sin x + \cos x}$. **(10 marks)** [MATH 145 | L-1/T-1 | 2020–21] [MATH 145 | L-1/T-1 | 2016–17]

Q3. Find $\int_0^\pi \frac{x}{\sin x + 1} dx$. **(11 marks)** [MATH 145 | L-1/T-1 | 2018–19]

Q4. Evaluate $\int_0^{\pi/2} \frac{\sin^2 x}{1 + \sin x \cos x} dx$. **(12 marks)** [MATH 145 | L-1/T-1 | 2017–18]

Q5. Find the value of $\int_0^\pi \frac{x dx}{a^2 - \cos^2 x}$. **(12½ marks)** [MATH 143 | L-1/T-2 | 2013–14]

Q6. Evaluate: $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$. **(12½ marks)** [MATH 143 | L-1/T-2 | 2012–13]

Q7. Find the value of $\int_0^\pi \frac{x \tan x}{\sec x + \tan x} dx$. **(11½ marks)** [MATH 143 | L-1/T-2 | 2011–12]

Definite Integrals: Substitution & Standard Methods

Q1. Evaluate $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$. **(12 marks)** [MATH 145 | L-1/T-1 | 2017–18]

Q2. Evaluate $\int_0^{\pi/2} x \ln(\sin x) dx$. **(12 marks)** [MATH 145 | L-1/T-1 | 2016–17]

Q3. Evaluate $\int_0^{\pi/2} \frac{\sin x \cos x}{a^2 \sin^2 x + b^2 \cos^2 x} dx$. **(11 marks)** [MATH 145 | L-1/T-1 | 2015–16]

Q4. Find the value of $\int_0^1 \frac{\sqrt{x(1-x)^2}}{1+x} dx$. **(11 marks)** [MATH 143 | L-1/T-2 | 2012–13]

Reduction Formula & Wallis' Formula

Q1. If $\int_0^{\pi/2} \sin^m x \cos^n x dx = \frac{m-1}{m+n} I_{m-2,n}$, evaluate $\int_0^{\pi/2} \sin^{10} x \cos^2 x dx$ by Wallis' formula, where $I_{m,n} = \int_0^{\pi/2} \sin^m x \cos^n x dx$. **(12 marks)** [MATH 141 | L-1/T-1 | 2024–25]

Q2. Find the reduction formula for $\int \sin^n x \cos^m x dx$ and hence evaluate $\int \sin^3 x \cos^2 x dx$. **(15 marks)** [MATH 141 | L-1/T-1 | 2023–24]

Q3. Find the reduction formula for $\int e^{ax} \cos^n x dx$. Also verify that the reduction formula for $\int \cos^n x dx$ is a special case. **(12 marks)** [MATH 141 | L-1/T-1 | 2022–23]

Q4. Find a reduction formula for $\int \sin nx \sec x dx$ and hence find $\int \sin 3x \sec x dx$. **(10 marks)** [MATH 141 | L-1/T-1 | 2021–22]

- Q5. Find Wallis' formula for $\int_0^{\pi/2} \sin^m x \cos^n x dx$. (12 marks) [MATH 145 | L-1/T-1 | 2018–19]
- Q6. Derive Wallis' formula for $\int_0^{\pi/2} \cos^n x dx$ and hence evaluate $\int_0^{\pi/2} \cos^9 x dx$. (11 marks) [MATH 145 | L-1/T-1 | 2016–17]
- Q7. Derive a reduction formula for $\int e^{ax} \cos^n x dx$. (12 marks) [MATH 143 | L-1/T-2 | 2013–14, 2011–12]
- Q8. Obtain the reduction formula for $\int \cos^n x dx$. (11 marks) [MATH 143 | L-1/T-2 |]

Improper Integrals

- Q1. Evaluate $\int_0^1 x^5 \sqrt{\frac{1+x^2}{1-x^2}} dx$. (12 marks) [MATH 141 | L-1/T-1 | 2022–23]
- Q2. Evaluate $\int_0^\infty \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}}, k^2 < 1$. (12 marks) [MATH 145 | L-1/T-1 | 2020–21]
- Q3. Evaluate $\int_0^\infty \frac{\ln\left(x + \frac{1}{x}\right)}{1+x^2} dx$. (10 marks) [MATH 141 | L-1/T-1 | 2021–22]
- Q4. Evaluate $\int_0^1 \frac{1}{\sqrt{x(1-x)}(1+x)(x+2)} dx$. (10 marks) [MATH 145 | L-1/T-1 | 2018–19] [MATH 143 | L-1/T-2 | 2014–15]
- Q5. Evaluate $\int_0^\infty \frac{\sqrt{x}}{(1+x)^2} dx$. (11 marks) [MATH 145 | L-1/T-1 | 2016–17]
- Q6. Show that $\int_0^\infty \frac{\ln(1+x^2)}{1+x^2} dx = \pi \ln 2$. (12 $\frac{2}{3}$ marks) [MATH 143 | L-1/T-2 | 2010–11]

Beta & Gamma Functions

- Q1. Evaluate the integral $\int_0^a y^4 \sqrt{a^2 - y^2} dy$ by the concept of the Gamma function. (10 marks) [MATH 141 | L-1/T-1 | 2024–25]
- Q2. Show that $\int_0^{\pi/2} \sin^p x \cos^q x dx = \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{2 \Gamma\left(\frac{p+q+2}{2}\right)}$. (10 marks) [MATH 141 | L-1/T-1 | 2023–24]
- Q3. Write the first and second Eulerian integrals. Hence prove $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$. (15 marks) [MATH 141 | L-1/T-1 | 2021–22]
- Q4. Prove that $\int_0^1 \frac{x^{m-1}(1-x)^{n-1}}{(a+x)^{m+n}} dx = \frac{\beta(m, n)}{a^n(a+1)^m}$. (11 marks) [MATH 145 | L-1/T-1 | 2020–21]
- Q5. Show that $\Gamma\left(n + \frac{1}{2}\right) = \frac{\Gamma(2n+1)\sqrt{\pi}}{2^{2n}\Gamma(n+1)}$. (15 marks) [MATH 145 | L-1/T-2 | 2019–20]
- Q6. Prove that $\int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - \frac{1}{2}\sin^2 \phi}} = \frac{[\Gamma(1/4)]^2}{4\sqrt{\pi}}$. (12 marks) [MATH 145 | L-1/T-1 | 2018–19] [MATH 145 | L-1/T-1 | 2015–16]
- Q7. Show that $\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{1}{2}\beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right); m > -1, n > -1$. (12 marks) [MATH 145 | L-1/T-1 | 2017–18]
- Q8. Prove that $\Gamma(1/2) = \sqrt{\pi}$. Hence evaluate $\int_0^\infty \sqrt{y} e^{-y^3} dy$. (13 marks) [MATH 145 | L-1/T-1 | 2017–18]
- Q9. Show that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$. (12 marks) [MATH 145 | L-1/T-1 | 2016–17] [MATH 143 | L-1/T-2 | 2011–12]

Q10. Evaluate: (i) $\int_0^\infty e^{-y^{1/n}} dy$ (ii) $\Gamma(-3/2)$. **(12 marks)**

[MATH 145 | L-1/T-1 | 2016–17]

Area Under Curves

Q1. Find the area of the region inside the curve $r = \sin \theta$ and outside the curve $r = 1 - \cos \theta$. **(12 marks)** [MATH 141 | L-1/T-1 | 2024–25] [MATH 145 | L-1/T-1 | 2015–16] [MATH 143 | L-1/T-2 | 2011–12]

Q2. Find the area of the region inside the circle $r = 3 \sin \theta$ and outside the cardioid $r = 1 + \sin \theta$. **(11 marks)** [MATH 141 | L-1/T-1 | 2023–24]

Q3. Use a polar double integral to find the area enclosed by the three-petaled rose $r = \sin 3\theta$. **(15 marks)** [MATH 141 | L-1/T-1 | 2022–23]

Q4. Find the area of the region inside the circle $r = 1$ and outside the cardioid $r = 1 + \cos \theta$. **(11 marks)** [MATH 141 | L-1/T-1 | 2021–22]

Q5. Calculate the area of the inner loop of the curve $r = b(-1 + 2 \cos \theta)$. **(11 marks)** [MATH 145 | L-1/T-1 | 2020–21]

Q6. Estimate the area of the region enclosed by $y^2 = 4x$ and $y = 2x - 4$. **(15 marks)** [MATH 145 | L-1/T-2 | 2019–20]

Q7. Find the area enclosed by the cardioid $r = a(1 + \sin \theta)$. **(12 marks)** [MATH 145 | L-1/T-1 | 2018–19]

Q8. Find the area of the region enclosed by the rose curve $r = \cos 2\theta$. **(10 marks)** [MATH 145 | L-1/T-1 | 2017–18]

Q9. Find the area bounded by the cardioid $r = c(1 + \cos \theta)$. **(11 marks)** [MATH 145 | L-1/T-1 | 2016–17]

Q10. Find the area between the curve $y^2(2a - x) = x^3$ and its asymptote. **(16½ marks)** [MATH 143 | L-1/T-2 | 2014–15]

Q11. Find the total area interior to $y^2 = 2ax - x^2$ and exterior to $y^2 = ax$ lying in the first quadrant. **(22 marks)** [MATH 143 | L-1/T-2 | 2013–14]

Q12. Find the area bounded by the curve $y^2(a + x) = (a - x)^3$ and its asymptote. **(16 marks)** [MATH 143 | L-1/T-2 | 2012–13]

Q13. Find the area interior to the curve $r = \cos \theta$ and exterior to the curve $r^2 = \cos 2\theta$. **(12 marks)** [MATH 143 | L-1/T-2 | 2010–11]

Arc Lengths

Q1. Define arc length of a curve. Find the arc length of the cardioid $r = a(1 + \cos \theta)$. **(10 marks)** [MATH 141 | L-1/T-1 | 2024–25]

Q2. Find the arc length of the curve $x = e^t(\sin t + \cos t)$, $y = e^t(\cos t - \sin t)$ for $1 \leq t \leq 4$. **(12 marks)** [MATH 141 | L-1/T-1 | 2023–24]

Q3. Find the exact arc length of the curve $x = \frac{1}{8}y^4 + \frac{1}{4}y^{-2}$ from $y = 1$ to $y = 4$. **(10 marks)** [MATH 141 | L-1/T-1 | 2022–23]

Q4. Find the arc length of the curve $r = \left(\sin \frac{\theta}{4}\right)^4$ for $\theta = 0$ to π . **(12 marks)** [MATH 145 | L-1/T-1 | 2020–21]

Q5. Show that the total length of the curve $x^2(a^2 - x^2) = 8a^2y^2$ is $\frac{\pi a}{\sqrt{2}}$. **(10 marks)** [MATH 145 | L-1/T-1 | 2018–19]

Q6. Find the exact arc length of the curve $24xy = y^4 + 48$ from $y = 2$ to $y = 4$. **(12 marks)** [MATH 145 | L-1/T-1 | 2017–18]

Q7. Find the length of the semi-cubical parabola $ay^2 = x^3$ from the vertex to the point (a, a) . **(12 marks)** [MATH 145 | L-1/T-1 | 2016–17]

Q8. Find the length of the perimeter of the asteroid $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{a}\right)^{2/3} = 1$. **(24½ marks)** [MATH 143 | L-1/T-2 | 2013–14]

Q9. Find the perimeter of the loop of the curve $9ay^2 = (x - 2a)(x - 5a)^2$. **(14½ marks)** [MATH 143 | L-1/T-2 | 2011–12]

Volume & Surface Area of Revolution

Q1. Use cylindrical shells to find the volume of the solid generated when the region R under $y = x^2$ over the interval $[0, 2]$ is revolved about the line $y = -1$. **(12 marks)** [MATH 141 | L-1/T-1 | 2024–25]

- Q2.** A solid has, as its base, the circular region in the xy -plane bounded by $x^2 + y^2 = r^2$ with $r > 0$. Find the volume of the solid if every cross-section by a plane perpendicular to the x -axis is an equilateral triangle with one side in the base. **(13 marks)** [MATH 141 | L-1/T-1 | 2024–25]
- Q3.** Find the surface area of the object formed by revolving the asteroid $x^{2/3} + y^{2/3} = a^{2/3}$ about the y -axis. **(12 marks)** [MATH 141 | L-1/T-1 | 2024–25]
- Q4.** Find the volume of the solid whose base is the region enclosed between $x = 1 - y^2$ and the y -axis, and whose cross-sections perpendicular to the y -axis are squares. **(11 marks)** [MATH 141 | L-1/T-1 | 2023–24]
- Q5.** The line segment $y = 1 - x$, $0 \leq y \leq 1$, is revolved about the y -axis to generate a cone. Find its lateral surface area. **(12 marks)** [MATH 141 | L-1/T-1 | 2023–24]
- Q6.** Find the volume of the solid generated when the region between $f(x) = \frac{1}{2} + x^2$ and $g(x) = x$ over $[0, 1]$ is revolved about the x -axis. **(15 marks)** [MATH 141 | L-1/T-1 | 2022–23]
- Q7.** Find the area of the surface generated by revolving $x = 2\sqrt{1-y}$, $0 \leq y \leq 2$, about the y -axis. **(10 marks)** [MATH 141 | L-1/T-1 | 2022–23]
- Q8.** Find the area of the surface generated by revolving $x^2 + y^2 = 9$ about the y -axis for $-2 \leq y \leq 2$. **(12 marks)** [MATH 141 | L-1/T-1 | 2021–22]
- Q9.** Find the surface area of the solid generated by revolving the portion of the catenary $y = \frac{a}{2}(e^{x/a} + e^{-x/a})$ from $x = 0$ to $x = a$ about the y -axis. **(12 marks)** [MATH 145 | L-1/T-1 | 2020–21]
- Q10.** Find the volume of the solid generated by revolving the cardioid $r = a(1 - \cos \theta)$ about the initial line. **(15 marks)** [MATH 145 | L-1/T-2 | 2019–20]
- Q11.** Find the volume of the solid formed by revolving the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$, about its minor axis. **(13 marks)** [MATH 145 | L-1/T-1 | 2018–19]
- Q12.** Find the volume of the solid formed by the revolution about the x -axis of a loop of the curve $(x - 4a)y^2 = ax(x - 3a)$. **(13 marks)** [MATH 145 | L-1/T-1 | 2017–18]
- Q13.** Find the surface area of the solid generated by revolving the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about its (i) major axis and (ii) minor axis. **(11 marks)** [MATH 145 | L-1/T-1 | 2015–16]
- Q14.** Find the volume of the solid generated by revolving the parabola $\sqrt{x} + \sqrt{y} = \sqrt{a}$, bounded by the axes $x = 0$, $y = 0$, about the x -axis. **(11 marks)** [MATH 145 | L-1/T-1 | 2015–16]
- Q15.** Find the surface area of the solid generated by revolving the asteroid $x^{2/3} + y^{2/3} = a^{2/3}$ about the x -axis. **(15 marks)** [MATH 143 | L-1/T-2 | 2014–15]
- Q16.** Find the volume and surface area of the solid generated by revolving the cardioid $r = a(1 + \cos \theta)$ about the initial line. **(30½ marks)** [MATH 143 | L-1/T-2 | 2012–13]
- Q17.** Find the area between the cissoid $y^2(2a - x) = x^3$ and its asymptote. Also find the volume and surface area of the solid formed by revolving the cissoid about its asymptote. **(24½ marks)** [MATH 143 | L-1/T-2 | 2010–11]

Multiple Integrals

- Q1.** Use a triple integral to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$. **(11 marks)** [MATH 141 | L-1/T-1 | 2024–25] [MATH 145 | L-1/T-2 | 2019–20]
- Q2.** Evaluate $\int_1^2 \int_{1/y}^y \sqrt{\frac{y}{x}} e^{\sqrt{xy}} dx dy$. **(12 marks)** [MATH 141 | L-1/T-1 | 2023–24]
- Q3.** Let G be the solid in the first octant bounded by $x^2 + y^2 + z^2 = 4$ and the coordinate planes. Evaluate $\iiint_G xyz dV$ using spherical coordinates. **(12 marks)** [MATH 141 | L-1/T-1 | 2023–24]
- Q4.** Use triple integration in cylindrical coordinates to find the volume of the solid bounded above by $z = \sqrt{25 - x^2 - y^2}$, below by the xy -plane, and laterally by $x^2 + y^2 = 9$. **(20 marks)** [MATH 141 | L-1/T-1 | 2022–23]
- Q5.** Use double integration to find the volume of the solid in the first octant bounded above by $z = 9 - x^2$, below by $z = 0$, and laterally by $y^2 = 3x$. **(12 marks)** [MATH 141 | L-1/T-1 | 2021–22]

- Q6.** Evaluate $\iiint_R \frac{1}{\sqrt{x+y}} dy dx dz$ where $R : x \leq y \leq z, 0 \leq x \leq z, 1 \leq z \leq 4$. **(11 marks)** [MATH 141 | L-1/T-1 | 2021–22]
- Q7.** Using spherical coordinates, find the volume of the region common to both the sphere $x^2 + y^2 + z^2 = 4$ and the cylinder $x^2 + y^2 = 1$. **(12 marks)** [MATH 145 | L-1/T-1 | 2020–21]
- Q8.** Evaluate $\int_0^2 \int_0^z \int_0^{x\sqrt{3}} \frac{x}{x^2 + y^2} dy dx dz$. **(13 marks)** [MATH 145 | L-1/T-1 | 2018–19]
- Q9.** Evaluate $\int_{0\text{sqrt}2}^x \int_0^x \int_0^{2-x^2} xyz dz dy dx$. **(10 marks)** [MATH 145 | L-1/T-1 | 2017–18]
- Q10.** Evaluate the triple integral $\int_0^2 \int_0^{\frac{1}{3\sqrt{4-x^2}}} \int_{x^2+y^2}^{4-8y^2} dz dy dx$. **(15 marks)** [MATH 143 | L-1/T-2 | 2014–15]
- Q11.** Use a double integral to find the volume of the tetrahedron bounded by the coordinate planes and the plane $z = 4 - 4x - 2y$. **(10 marks)** [MATH 143 | L-1/T-2 | 2010–11]

BUET · MATH 141 Question Bank

Ordinary Differential Equations

First Order, Higher Order, Euler's Equations & more

S3. Ordinary Differential Equations

Formation of Differential Equations

- Q1.** Obtain the differential equation of the family of circles tangent to the x -axis and sketch several representative members of the family. **(11 marks)** [MATH 141 | L-1/T-1 | 2024–25]
- Q2.** Define degree and order of an ODE with an example. Find the degree and order of the DE corresponding to the family of curves $y = c(x - c)^2$, where c is an arbitrary constant. **(10 marks)** [MATH 141 | L-1/T-1 | 2023–24] [MATH 147 | L-1/T-2 | 2015–16]
- Q3.** Find the differential equation of the family of curves $y = Ae^{3x} + Be^{5x}$ for different values of A and B . **(11 marks)** [MATH 141 | L-1/T-1 | 2021–22]
- Q4.** Find the differential equation of all circles. **(15 marks)** [MATH 147 | L-1/T-2 | 2017–18]
- Q5.** Find the differential equation of all circles of radius a . **(10½ marks)** [MATH 143 | L-1/T-2 | 2011–12, 2010–11]

First Order ODEs

- Q1.** Define exact differential equation. Determine whether the following differential equation is exact, and hence solve it:

$$\frac{dy}{dx} = \frac{xy^2 - \cos x \sin x}{y(1 - x^2)}, \quad y(0) = 2.$$

(13 marks)

[MATH 141 | L-1/T-1 | 2024–25]

- Q2.** Check the linearity of the differential equation $\frac{dy}{dx} + \frac{y}{x-1} = xy^{1/3}$ and find the general solution. **(11 marks)** [MATH 141 | L-1/T-1 | 2024–25]
- Q3.** State Bernoulli's equation. Solve $\frac{dy}{dx} + \frac{xy}{1-x^2} = x\sqrt{y}$ when $y(0) = 0$. **(13 marks)** [MATH 141 | L-1/T-1 | 2023–24]
- Q4.** Consider $(2y + 3xy^2) dx + (x + 2x^2y) dy = 0$. Show it is not exact. Find an integrating factor and solve the resulting exact equation. **(12 marks)** [MATH 141 | L-1/T-1 | 2023–24]
- Q5.** Sketch the slope field of $\frac{dy}{dx} = \frac{x}{y^2}$ and show the solution passing through $y(0) = 1$. **(10 marks)** [MATH 141 | L-1/T-1 | 2022–23]
- Q6.** Determine whether $(2y \sin x \cos x - y + 2y^2 e^{xy^2}) dx - (x - \sin^2 x - 4xye^{xy^2}) dy = 0$ is exact. If not, make it exact and solve. **(15 marks)** [MATH 141 | L-1/T-1 | 2022–23]
- Q7.** Solve the initial value problem $(2x - 5y) dx + (4x - y) dy = 0$, $y(1) = 6$. **(13 marks)** [MATH 141 | L-1/T-1 | 2021–22]
- Q8.** Solve: $x(x-1)\frac{dy}{dx} - (x-2)y = \frac{x^4}{\sqrt{x^2-1}}$ when $y(2) = 4$. **(11 marks)** [MATH 141 | L-1/T-1 | 2021–22]
- Q9.** Consider $(y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0$. Show it is not exact. Find an integrating factor of the form $f(y)$ and solve the resulting exact equation. **(11 marks)** [MATH 141 | L-1/T-1 | 2021–22]
- Q10.** Find an equation of the curve that has slope $(6x+1)y^2 \frac{dy}{dx} + 3x^2 + 2y^3 = 0$ and passes through $(0, 0)$. **(16.67 marks)** [MATH 147 | L-1/T-2 | 2020–21]
- Q11.** Find the curve y satisfying $3(1+t^2)\frac{dy}{dt} = 2ty(y^3 - 1)$, $y(0) = 1$. **(15 marks)** [MATH 147 | L-1/T-2 | 2020–21]
- Q12.** Solve: $\frac{dy}{dx} + \frac{3}{x}y = 27y^{1/3} \ln x$, $x > 0$. **(16 marks)** [MATH 147 | L-1/T-2 | 2019–20]
- Q13.** Solve: $y(y^2 - 2x^2) dx + x(2y^2 - x^2) dy = 0$. **(20 marks)** [MATH 147 | L-1/T-2 | 2019–20]
- Q14.** Solve: $x(1-x^2)\frac{dy}{dx} + y(2x^2 - 1) = ax^3$. **(20 marks)** [MATH 147 | L-1/T-2 | 2018–19]
- Q15.** Solve: $x(x-1)\frac{dy}{dx} - y = x^2(x-1)^2$. **(15 marks)** [MATH 147 | L-1/T-2 | 2017–18]

- Q16. Solve: $\frac{dy}{dx} - y \sec x = y^2 \sin x \cos x$. (13 marks) [MATH 147 | L-1/T-2 | 2016–17]
- Q17. Solve: $\left(x \tan \frac{y}{x} + y\right) dx - x dy = 0$. (15 marks) [MATH 147 | L-1/T-2 | 2015–16]
- Q18. Solve the initial value problem: $x \frac{dy}{dx} - 2y = 2x^4$, $y(2) = 8$. (15 marks) [MATH 147 | L-1/T-2 | 2015–16]
- Q19. Solve: $\left(x^3 + y^2 \sqrt{x^2 + y^2}\right) dx = xy \sqrt{x^2 + y^2} dy$. (14 marks) [MATH 143 | L-1/T-2 | 2014–15]
- Q20. Solve: $\frac{dy}{dx} + \frac{y \ln y}{x} = \frac{y(\ln y)^2}{x^2}$. (16 marks) [MATH 143 | L-1/T-2 | 2014–15]
- Q21. Solve: $\frac{dy}{dx} = \frac{3y - 7x + 7}{3x - 7y - 3}$. (17½ marks) [MATH 143 | L-1/T-2 | 2013–14]
- Q22. Solve the initial value problem: $\left(y + \sqrt{x^2 + y^2}\right) dx - x dy = 0$, $y(1) = 0$. (15 marks) [MATH 143 | L-1/T-2 | 2013–14] [MATH 143 | L-1/T-2 | 2010–11]
- Q23. Solve: $\frac{dy}{dx} + \frac{1}{x} \sin 2y = x^3 \cos^2 y$. (14 marks) [MATH 143 | L-1/T-2 | 2013–14]
- Q24. Solve: $(2x - 5y + 3) dx - (2x + 4y - 6) dy = 0$. (16½ marks) [MATH 143 | L-1/T-2 | 2012–13]
- Q25. Solve: $3xy^2(1 - x^2) \frac{dy}{dx} + (2x^2 - 1)y = ax^3$. (16 marks) [MATH 143 | L-1/T-2 | 2012–13]
- Q26. Solve: $(2xy^4 e^y + 2xy^3 + y) dx + (x^2 y^4 e^y - x^2 y^2 - 3x) dy = 0$. (14 marks) [MATH 143 | L-1/T-2 | 2012–13]
- Q27. Solve the following:
- (a) $x \cos\left(\frac{y}{x}\right) (y dx + x dy) = y \sin\left(\frac{y}{x}\right) (x dy - y dx)$
- (b) $\frac{dy}{dx} + \frac{x - y - 2}{x - 2y - 3} = 0$
- (c) $(1 + y^2) dx = (\tan^{-1} y - x) dy$
- (36 marks) [MATH 143 | L-1/T-2 | 2011–12]
- Q28. Solve:
- (a) $\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$.
- (b) $y^2 dx + (3xy - 1) dy = 0$
- (12+12 marks) [MATH 143 | L-1/T-2 | 2010–11]

Second & Higher Order Linear ODEs

- Q1. Define fundamental set of solution. Find the general solution of
- $$y'' - 2y' + y = 10e^{-2x} \cos x.$$
- (12 marks) [MATH 141 | L-1/T-1 | 2024–25]
- Q2. Apply the method of variation of parameters to solve $(D^2 + 2D + 2)y = e^{-x} \csc x$. (12 marks) [MATH 141 | L-1/T-1 | 2023–24]
- Q3. Solve $\frac{d^2 y}{dx^2} + y = e^x \sin x$. (8 marks) [MATH 141 | L-1/T-1 | 2022–23]
- Q4. Solve $\frac{d^2 y}{dx^2} + 9y = x \sin 2x$. (12 marks) [MATH 141 | L-1/T-1 | 2021–22]
- Q5. Apply the method of variation of parameters to solve $\frac{d^2 y}{dx^2} + 4y = \sec 2x$. (11 marks) [MATH 141 | L-1/T-1 | 2021–22]
- Q6. Find a differential operator that annihilates $f(x) = 7x^4 + 7 + 3xe^{-3x} + 3 \cos 2x$ and hence solve $y'' + 8y' = f(x)$. (16½ marks) [MATH 147 | L-1/T-2 | 2020–21]
- Q7. Solve $\frac{d^2 y}{dx^2} - y = \cosh x \cos x$ using an appropriate technique. (15 marks) [MATH 147 | L-1/T-2 | 2020–21, 2016–17]

- Q8.** Solve $\frac{d^2y}{dx^2} + \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^3 = 0$. **(15 marks)** [MATH 147 | L-1/T-2 | 2020–21]
- Q9.** Find the complementary function and particular integral of: $\frac{d^2y}{dx^2} - y = xe^x + \cos^2 x$. **(20 marks)** [MATH 147 | L-1/T-2 | 2019–20]
- Q10.** Solve: $\frac{d^2y}{dx^2} + y = e^x \sin x + xe^{2x}$. **(20 marks)** [MATH 147 | L-1/T-2 | 2018–19]
- Q11.** Solve the differential equation $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$. **(15 marks)** [MATH 147 | L-1/T-2 | 2017–18]
- Q12.** Solve: $(D^4 + 2D^2 + 1)y = x^2 \cos x$, where $D = \frac{d}{dx}$. **(15 marks)** [MATH 147 | L-1/T-2 | 2017–18] [MATH 143 | L-1/T-2 | 2010–11]
- Q13.** Solve: $\frac{d^2y}{dx^2} + a^2y = \frac{a^2R}{p}(l - x)$ subject to $y = 0$ and $\frac{dy}{dx} = 0$ at $x = 0$. **(16 marks)** [MATH 147 | L-1/T-2 | 2017–18]
- Q14.** Find a differential operator that annihilates $f(x) = 7x^4 + 7 + 3xe^{-3x} + 3\cos 2x$ and hence solve $y''' + 8y'' = f(x)$. **(18 marks)** [MATH 147 | L-1/T-2 | 2016–17]
- Q15.** Find the general solution of $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10\sin x$. **(15 marks)** [MATH 147 | L-1/T-2 | 2015–16]
- Q16.** Solve: $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = x^3$. **(15 marks)** [MATH 147 | L-1/T-2 | 2015–16]
- Q17.** Solve $(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x$. **(14 marks)** [MATH 143 | L-1/T-2 | 2014–15]
- Q18.** Solve $(x+1)\frac{d^2y}{dx^2} + (x-1)\frac{dy}{dx} - 2y = 0$ by the method of factorization of operators. **(16 marks)** [MATH 143 | L-1/T-2 | 2014–15]
- Q19.** Find the general solution of $(D^2 + 16)y = \sec 4x$. **(15 marks)** [MATH 143 | L-1/T-2 | 2013–14]
- Q20.** Find the general solution of $[(2x-3)^2D^2 - 6(2x-3)D + 12]y = x^2 + x$. **(15 marks)** [MATH 143 | L-1/T-2 | 2013–14]
- Q21.** Find the general solution of $[(x+3)D^2 - (2x+7)D + 2]y = (x+3)^2e^x$. **(16 marks)** [MATH 143 | L-1/T-2 | 2013–14]
- Q22.** Find the general solution of $(D^3 - 7D - 6)y = e^{2x} \cdot x^2$. **(16 marks)** [MATH 143 | L-1/T-2 | 2012–13]
- Q23.** Find the general solution of $(D^2 - 6D + 13)y = \sin 2x \cdot e^{3x}$. **(14 marks)** [MATH 143 | L-1/T-2 | 2012–13]
- Q24.** Find the general solution of $(D^2 + 2D + 1)y = x \cos x$. **(14 marks)** [MATH 143 | L-1/T-2 | 2011–12]
- Q25.** Solve $[xD^2 + (x-1)D - 1]y = x^2$ by the method of factorization of operators. **(16 marks)** [MATH 143 | L-1/T-2 | 2011–12]
- Q26.** Find the general solution of $y(1 - \ln y)\frac{d^2y}{dx^2} + (1 + \ln y)\left(\frac{dy}{dx}\right)^2 = 0$. **(14 marks)** [MATH 143 | L-1/T-2 | 2010–11]

Euler's Homogeneous Linear Equations

- Q1.** Solve the differential equation $(x^3D^3 - 4x^2D^2 + 8xD - 8)y = 4\ln x$. **(11 marks)** [MATH 141 | L-1/T-1 | 2024–25]
- Q2.** State Cauchy-Euler equation. Solve $[(3x+2)^2D^2 + 3(3x+2)D - 36]y = x^2 + x$. **(11 marks)** [MATH 141 | L-1/T-1 | 2023–24]
- Q3.** Discuss the Cauchy-Euler equation. Solve $x^2y'' + xy' - y = \frac{1}{x+1}$. **(15 marks)** [MATH 141 | L-1/T-1 | 2022–23] [MATH 147 | L-1/T-2 | 2018–19]
- Q4.** Reduce $x^3\frac{d^3y}{dx^3} + 3x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = x \log x$ to linear form with constant coefficients and then solve. **(12 marks)** [MATH 141 | L-1/T-1 | 2021–22]
- Q5.** Reduce the Cauchy-Euler equation $x^2\frac{d^2y}{dx^2} + 10x\frac{dy}{dx} + \frac{8y}{x} = x$ to constant coefficients and solve. **(16 $\frac{2}{3}$ marks)** [MATH 147 | L-1/T-2 | 2020–21]

- Q6. Solve: $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 5y = x^2 \sin(\ln x)$. (15 marks) [MATH 147 | L-1/T-2 | 2020–21] [MATH 147 | L-1/T-2 | 2016–17]
- Q7. Solve: $x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = x \ln x$. (26 marks) [MATH 147 | L-1/T-2 | 2019–20]
- Q8. Solve: $(x^2 D^2 - 2xD + 2)y = x^2 + \sin(5 \ln x)$. (16 marks) [MATH 147 | L-1/T-2 | 2017–18]
- Q9. Reduce the Cauchy-Euler equation $x \frac{d^2 y}{dx^2} + 10 \frac{dy}{dx} + \frac{8y}{x} = x$ to constant coefficients and hence solve. (18 marks) [MATH 147 | L-1/T-2 | 2016–17]
- Q10. Solve $2x^2 y'' - xy' + y = 4 \sin(\ln x)$. (16 marks) [MATH 143 | L-1/T-2 | 2014–15]
- Q11. Find the general solution of $(x^3 D^3 + 6x^2 D^2 + 8xD + 2)y = x^2 + \log x$. (16 marks) [MATH 143 | L-1/T-2 | 2012–13]
- Q12. Find the general solution of $(x^4 D^4 + 6x^3 D^3 + 9x^2 D^2 + 3xD + 1)y = (1 + \log x)^2$. (16 marks) [MATH 143 | L-1/T-2 | 2011–12]
- Q13. Find the general solution of $(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \ln(1+x)$. (16½ marks) [MATH 143 | L-1/T-2 | 2010–11]

Equations of First Order but Higher Degree

- Q1. Solve: $2y \frac{d^2 y}{dx^2} - 3 \left(\frac{dy}{dx} \right)^2 - 4y^2 = 0$. (20 marks) [MATH 147 | L-1/T-2 | 2018–19]
- Q2. Solve: $x \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - \frac{dy}{dx} = 0$. (15 marks) [MATH 147 | L-1/T-2 | 2017–18]
- Q3. Solve: $y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = y^2$. (15 marks) [MATH 147 | L-1/T-2 | 2017–18]
- Q4. Solve: $y \frac{d^2 y}{dx^2} - \left(\frac{dy}{dx} \right)^2 = y^2 \ln y$. (15 marks) [MATH 147 | L-1/T-2 | 2016–17]
- Q5. Solve:
- (a) $y \frac{d^2 y}{dx^2} - \left(\frac{dy}{dx} \right)^2 = y^3 \log y$. (16.67 marks)
- (b) $\frac{d^4 y}{dx^4} - \cot x \frac{d^3 y}{dx^3} = 0$ (15 marks) [MATH 147 | L-1/T-2 | 2015–16]
- Q6. Solve: $(xp + y)^2 = xy$, where $p = \frac{dy}{dx}$. (15 marks) [MATH 147 | L-1/T-2 | 2015–16]

Applications of ODEs

- Q1. A series circuit has a constant EMF of 100 V, a resistor of 10Ω , and a capacitor of 2×10^{-4} farads. The switch is closed at time $t = 0$, and the charge on the capacitor at this instant is zero. Find the charge and the current at time $t > 0$. (12 marks) [MATH 141 | L-1/T-1 | 2024–25]
- Q2. An 8-pound weight stretches a spring 2 feet. A damping force numerically equal to 2 times the instantaneous velocity acts on the system. Determine the equation of motion if released from equilibrium with an upward velocity of 3 ft/sec. Also interpret the result. (12 marks) [MATH 141 | L-1/T-1 | 2023–24]
- Q3. A 200-volt EMF is applied to an RC-series circuit with resistance 1000Ω and capacitance 5×10^{-6} F. Find the charge $q(t)$ on the capacitor if $i(0) = 0.4$. Determine charge and current at $t = 0.05$ sec. (10 marks) [MATH 141 | L-1/T-1 | 2022–23]
- Q4. A mass of 1 slug attached to a spring stretches it 2 feet. Starting at $t = 0$, external force $f(t) = e^{-t} \sin 4t$ is applied. The damping force is 8 times the instantaneous velocity. Find the equation of motion and analyse displacement as $t \rightarrow \infty$. (12 marks) [MATH 141 | L-1/T-1 | 2022–23]
- Q5. An object weighing 48 lb is released from rest at the top of a plane metal slide inclined at 30° to the horizontal. Air resistance equals half the velocity, and the coefficient of friction is one-quarter. Find the velocity 4 sec after release. If the slide is 24 ft long, find the velocity at the bottom. (13 marks) [MATH 141 | L-1/T-1 | 2021–22]

- Q6.** An RCL circuit in series has $R = 180 \Omega$, $C = \frac{1}{280} \text{ F}$, $L = 30 \text{ H}$, and applied voltage $E(t) = 10 \sin t$. With no initial charge but initial current 1 A at $t = 0$, derive the differential equation and find the subsequent charge on the capacitor. **(13 marks)** [MATH 147 | L-1/T-2 | 2020–21, 2016–17]
- Q7.** A thermometer is taken from an inside room to the outside where the air temperature is 5°F . After 1 minute it reads 55°F , and after 5 minutes it reads 30°F . Derive the temperature profile and compute the initial room temperature. **(15 marks)** [MATH 147 | L-1/T-2 | 2020–21]
- Q8.** A tank initially contains 20 L of water. A solution containing 1 g/L of chemical flows in at 3 L/min and the mixture flows out at 2 L/min. (i) Set up and solve the IVP for the amount of chemical at time t . (ii) When does the concentration reach 0.5 g/L? **(30 marks)** [MATH 147 | L-1/T-2 | 2019–20]
- Q9.** A 4 ft spring measures 8 ft after a mass weighing 8 lb is attached. The damping force is $\sqrt{2}$ times the instantaneous velocity. (i) Find the equation of motion if released from equilibrium with downward velocity 5 ft/s. (ii) Find the time and position of extreme displacement. **(26 marks)** [MATH 147 | L-1/T-2 | 2019–20]
- Q10.** A tank contains 200 litres of fluid in which 30 grams of salt is dissolved. Brine containing 1 gram of salt per litre is pumped into the tank at 4 L/min; the well-mixed solution is pumped out at the same rate. Find the number of grams of salt in the tank at time t . **(20 marks)** [MATH 147 | L-1/T-2 | 2018–19]
- Q11.** A mass weighing 4 lb is attached to a spring with constant 2 lb/ft. The medium offers a damping force numerically equal to the instantaneous velocity. The mass is released from 1 ft below equilibrium with downward velocity 8 ft/s. Determine: (i) the time at which the mass passes through the equilibrium position; (ii) the time at which the mass attains its extreme displacement. **(20 marks)** [MATH 147 | L-1/T-2 | 2018–19]
- Q12.** A certain culture of bacteria grows at a rate proportional to its size. If the size doubles in 4 days, find the time required for the culture to increase to 10 times its original size. **(16 marks)** [MATH 147 | L-1/T-2 | 2017–18]
- Q13.** Suppose a student carrying a flu virus returns to an isolated college campus of 1000 students. The rate at which the virus spreads is proportional to both the number x of infected students and the number not infected. Determine the number of infected students after 6 days if $x(4) = 50$. **(15 marks)** [MATH 147 | L-1/T-2 | 2016–17]
- Q14.** A capacitor of $\frac{2}{1010} \text{ F}$, a resistance of 1Ω and an inductance of $\frac{1}{20} \text{ H}$ are connected in series. At $t = 0$, $i = 0$ and charge on the capacitor is 1 Coulomb. Find the charge and current at $t = 0.01 \text{ s}$. **(16 marks)** [MATH 147 | L-1/T-2 | 2015–16]